UNIT-1 : INTRODUCTION

<table>
<tr><th>Sr. No.</th><th>Question</th></tr>
<tr><td>1</td><td>A small online clothing business uses a Generative AI tool to create product descriptions automatically.Apply the concept of Generative AI to demonstrate how it can improve operational efficiency and reduce workload in this business.</td></tr>
<tr><td>2</td><td>A hospital uses an AI chatbot to handle patient queries. The data is shown below:
<table>
<tr><th>Day</th><th>Queries Received</th><th>Queries Resolved by AI</th></tr>
<tr><td>Mon</td><td>500</td><td>420</td></tr>
<tr><td>Tue</td><td>600</td><td>510</td></tr>
<tr><td>Wed</td><td>550</td><td>500</td></tr>
</table>
Explain how Generative AI improves patient communication based on the data.</td></tr>
<tr><td>3</td><td>A startup uses AI tools to generate social media posts daily.Analyze how Generative AI influences marketing strategies in terms of personalization, audience targeting, and content effectiveness.</td></tr>
<tr><td>4</td><td>A teacher uses AI to generate quizzes for students.Apply the concept of Generative AI to demonstrate how it can be used to create personalized learning assessments.</td></tr>
<tr><td>5</td><td>A bank uses AI-generated reports for financial analysis.
<table>
<tr><th>Month</th><th>Manual Report Time (hrs)</th><th>AI Report Time (hrs)</th></tr>
<tr><td>Jan</td><td>20</td><td>8</td></tr>
<tr><td>Feb</td><td>18</td><td>7</td></tr>
</table>
Illustrate how Generative AI improves decision-making efficiency.</td></tr>
<tr><td>6</td><td>A YouTube creator uses AI to generate video scripts.
Apply the concept of Generative AI to demonstrate how it can reduce content creation time and improve productivity.</td></tr>
<tr><td>7</td><td>An e-commerce company uses AI chatbots for customer support.Apply the concept of Generative AI to show how AI chatbots can be used to improve customer satisfaction.</td></tr>
<tr><td>8</td><td>A software company uses AI to generate code snippets.Apply the concept of Generative AI to demonstrate how it enhances software development productivity.</td></tr>
<tr><td>9</td><td>A news agency uses AI to summarize articles.Apply the concept of Generative AI to show how it can improve information processing and content delivery.</td></tr>
<tr><td>10</td><td>A company uses AI tools to design logos.Apply the concept of Generative AI to demonstrate how it can be used to enhance creativity in design processes.</td></tr>
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<table>
<tr><td>11</td><td>A travel agency uses AI to generate personalized travel plans.
Apply the concept of Generative AI to show how it can enhance customer experience through personalization.</td></tr>
<tr><td>12</td><td>A YouTube creator uses Generative AI tools to automatically generate video scripts for their content.
(a) In what ways does Generative AI help in reducing the time required for content creation?
(b) Examine how the use of Generative AI improves the creator’s productivity.</td></tr>
<tr><td>13</td><td>A bank uses AI-based systems to detect and prevent fraudulent activities across its digital banking services. Analyze the effectiveness of Generative AI in improving the accuracy and efficiency of fraud detection compared to traditional methods.</td></tr>
<tr><td>14</td><td>A software company uses Generative AI tools to automatically generate code snippets for application development tasks.
(a) Demonstrate the use of Generative AI in generating code snippets for software development tasks.
(b) Illustrate how Generative AI improves productivity in the software development process.</td></tr>
<tr><td>15</td><td>A bank uses AI-based systems to detect and prevent fraudulent activities across its digital banking services. Analyze the effectiveness of Generative AI in improving the accuracy and efficiency of fraud detection compared to traditional methods.</td></tr>
<tr><td>16</td><td>A travel agency adopts Generative AI to design customized travel experiences.
Apply how Generative AI can be used to create personalized travel plans based on individual customer preferences, and explain how this improves the overall customer experience with an example.</td></tr>
<tr><td>17</td><td>A bank uses AI-based systems to detect and prevent fraudulent activities across its digital banking services. Analyze the effectiveness of Generative AI in improving the accuracy and efficiency of fraud detection compared to traditional methods.</td></tr>
<tr><td>18</td><td>A bank uses AI to detect fraud.
<table>
<tr><th>Year</th><th>Fraud Cases Detected (Manual)</th><th>Fraud Cases Detected (AI)</th></tr>
<tr><td>2022</td><td>120</td><td>210</td></tr>
<tr><td>2023</td><td>140</td><td>300</td></tr>
</table>
Analyze the effectiveness of AI in fraud detection.</td></tr>
<tr><td>19</td><td>A publishing company uses AI to write short stories. Analyze the impact on the publishing industry.</td></tr>
<tr><td>20</td><td>A hospital uses AI to generate patient reports.Discuss reliability concerns.</td></tr>
<tr><td>21</td><td>A company uses Large Language Models for customer service automation. Analyze scalability benefits.</td></tr>
<tr><td>22</td><td>A university integrates AI tools for research paper drafting.Analyze academic integrity issues.</td></tr>
</table>

<table>
<tr><td>23</td><td>A government uses AI tools to generate policy drafts.Examine benefits and risks.</td></tr>
<tr><td>24</td><td>A company uses AI for hiring.
<table><tr><th>Candidates</th><th>Selected (Manual)</th><th>Selected (AI)</th></tr><tr><td>Male</td><td>60</td><td>80</td></tr><tr><td>Female</td><td>40</td><td>20</td></tr></table>Analyze ethical challenges and bias.</td></tr>
<tr><td>25</td><td>A social media platform uses AI recommendations.Analyze impact on user behavior.</td></tr>
<tr><td>26</td><td>A company adopts AI for process automation.Explain efficiency improvements.</td></tr>
<tr><td>27</td><td>A legal firm uses AI to draft contracts.Analyze risks.</td></tr>
<tr><td>28</td><td>A company uses AI to generate product designs.Analyze impact on innovation.</td></tr>
<tr><td>29</td><td>A startup uses AI for forecasting.
<table><tr><th>Quarter</th><th>Actual Sales</th><th>AI Predicted</th></tr><tr><td>Q1</td><td>50000</td><td>48000</td></tr><tr><td>Q2</td><td>60000</td><td>62000</td></tr></table>Evaluate prediction accuracy.</td></tr>
<tr><td>30</td><td>A company uses AI voice assistants.Analyze impact on user interaction.</td></tr>
<tr><td>31</td><td>A healthcare company uses AI diagnosis suggestions. Discuss ethical concerns.</td></tr>
<tr><td>32</td><td>A company faces inaccurate AI outputs.Analyze limitations of Generative AI.</td></tr>
<tr><td>33</td><td>A government uses AI for public communication. Evaluate societal impact.</td></tr>
<tr><td>34</td><td>A company replaces human writers with AI.Evaluate economic impact.</td></tr>
<tr><td>35</td><td>AI-generated deepfakes spread misinformation:</td></tr>
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<tr><td></td><td>

Platform	Fake Videos Detected	Removal Rate (%)
A	500	60
B	800	40

Analyze ethical issues and propose regulations.</td></tr>
<tr><td>36</td><td>A healthcare system relies on AI diagnosis. Evaluate risks and safeguards.</td></tr>
<tr><td>37</td><td>AI model shows bias:

Group	Approval Rate (%)
Group A	80
Group B	45

Analyze bias and suggest mitigation strategies.</td></tr>
<tr><td>38</td><td>AI is used in global policy-making.Evaluate if AI can replace human judgment.</td></tr>
<tr><td>39</td><td>A business uses AI without transparency.Analyze need for explainable AI.</td></tr>
<tr><td>40</td><td>Workplace AI adoption:

Department	Human Work (%)	AI Work (%)
HR	70	30
Marketing	40	60
IT	30	70

Recommend a framework to balance AI and human roles.</td></tr>
</table>

UNIT -2 : Generative AI in Finance and Business

Sr. No.	Question
1	A fintech startup develops a trading system using GAN-generated synthetic stock data due to limited real data. Apply how synthetic data improves model performance and robustness.

2	A bank generates synthetic fraud transaction patterns due to limited real fraud data. Apply how this improves detection accuracy and training.
3	A firm uses Generative AI to simulate stock price scenarios instead of traditional models. Apply how this improves financial forecasting.
4	A bank deploys an LLM-based chatbot for handling customer queries. Apply how this improves service efficiency and user experience.
5	A marketing agency generates personalized ads using customer data. Apply how Generative AI improves creativity and personalization.
6	An e-commerce company simulates customer behavior during sales using AI. Apply how simulation improves decision-making.
7	A marketing company uses Generative AI to automatically create advertisements based on historical customer data. However, the training data contains inherent biases. Analyze how bias in Generative AI–generated advertisements can impact marketing outcomes such as audience targeting, brand perception, and campaign effectiveness. Support your answer with a suitable example**.**
8	A global logistics firm wants to predict and handle disruptions such as port delays and sudden demand spikes. Using Generative AI, analyze how synthetic scenario generation can help simulate disruptions and improve supply chain resilience. Explain with an example.
9	A marketing company uses Generative AI to automatically create advertisements based on historical customer data. However, the training data contains inherent biases. Analyze how bias in Generative AI–generated advertisements can impact marketing outcomes such as audience targeting, brand perception, and campaign effectiveness. Support your answer with a suitable example**.**
10	A trading platform simulates market conditions using AI. Apply how this optimizes trading strategies.
11	A trading platform leverages Generative AI to create simulated market scenarios such as bullish, bearish, and highly volatile conditions. (a) Apply how Generative AI can be used to generate realistic synthetic market data and trading environments to enhance and optimize trading strategies. (b) Support your answer with an example demonstrating improved trading decisions.
12	A logistics company simulates supply chain scenarios using AI. Apply how this improves operational efficiency.
13	An organization uses AI to automatically generate financial summaries for stakeholders based on large datasets. (a) Analyze the challenges related to accuracy in AI-generated financial summaries for stakeholders. (b) Examine the issues of accountability and responsibility when financial summaries are generated using AI systems.
14	A company automates invoice processing using AI. Apply how this reduces errors and improves accuracy.

15	A financial institution generates synthetic loan data. Apply how this improves credit risk modeling.
16	A business analyst simulates market conditions using AI. Apply how this supports strategic decision-making.
17	A hedge fund trains trading models using synthetic market scenarios. Analyze risks of overfitting to synthetic data.
18	A bank uses synthetic fraud patterns for detection. Analyze risks against evolving cyber threats.
19	A chatbot provides financial advice using AI. Analyze ethical risks in automated decision-making.
20	AI-generated ads use biased historical data. Analyze impact of bias on marketing outcomes.
21	AI generates financial reports with errors. Analyze risks of hallucinated outputs.
22	A firm uses Generative AI to automate the generation of financial reports from business data. Apply the concept of Generative AI to explain its use in automating financial report generation and improving productivity in the organization.
23	A business analyst uses AI tools to simulate market conditions for evaluating business strategies and improving decision-making. (a) Use AI simulation concepts to determine the purpose of simulating market conditions. (b) Illustrate two benefits of using AI in improving business decision-making.
24	AI-generated ads use biased historical data. Analyze impact of bias on marketing outcomes.
25	A chatbot struggles with complex financial queries. Analyze its limitations.
26	A fraud system uses real and synthetic data. Analyze trade-offs in training.
27	AI generates financial summaries for stakeholders. Analyze challenges in accuracy and accountability.
28	AI trained on one country’s data is used globally. Analyze impact on fairness and performance.
29	AI simulates competitor strategies. Analyze reliability and limitations.
30	A forecasting model ignores rare events. Analyze impact on prediction accuracy and risk.
31	AI is used for hiring decisions. Analyze bias and fairness issues.

32	A chatbot struggles with multilingual users. Analyze scaling challenges.
33	A trading firm compares real vs synthetic data models using a given dataset.
34	A bank trains fraud models using synthetic data with a given dataset.
35	A chatbot shows lower performance for complex queries (dataset given).

33 A trading firm compares real vs synthetic data models using a given dataset.

Model Type		Accuracy (%)	Profit (%)	Risk Score
Real Data Model		88	12	0.3
Synthetic Data Model		82	15	0.5

Market Conditions

Scenario	Volatility	Model Loss (%)
Stable Market	Low	5
Volatile Market	High	20

Evaluate effectiveness and suggest improvements.

34 A bank trains fraud models using synthetic data with a given dataset.

Dataset

Feature	Real Data	Synthetic Data
Fraud Rate (%)	3	5
Transaction Variance	High	Medium
Detection Accuracy (%)	92	85

Risk Metrics

Metric	Value
False Positives	12%
False Negatives	8%

Propose validation strategies.

35 A chatbot shows lower performance for complex queries (dataset given).

	Dataset | Query Type | Accuracy (%) | Response Time (sec) | Satisfaction Score | | Simple Queries | 95 | 2 | 4.5 | | Complex Queries | 70 | 5 | 3.2 | Recommend improvements.

Dataset

Query Type	Accuracy (%)	Response Time (sec)	Satisfaction Score
Simple Queries	95	2	4.5
Complex Queries	70	5	3.2

Recommend improvements.

36 AI vs human marketing campaigns are compared (dataset given).

Dataset

Campaign	Engagement (%)	Conversion Rate (%)	ROI (%)
Human Created	65	10	15
AI Generated	75	12	18

Risk Factors

Issue	Impact
Bias in Ads	High
Lack of Creativity	Medium

Evaluate benefits and propose hybrid strategies.

37 A forecasting model shows prediction errors (dataset given).

Dataset

Year	Actual Revenue	Predicted Revenue	Error (%)
2022	100 Cr	95 Cr	5
2023	120 Cr	140 Cr	16

Analyze errors and suggest improvements.

38 AI-based loan approval decisions are given (dataset).

	Dataset
	Evaluate fairness and reliability.

Dataset

Customer ID	Credit Score	AI Risk Score	Loan Approved (Y/N)
C1	750	0.2	Y
C2	600	0.6	N
C3	680	0.4	Y

Evaluate fairness and reliability.

39 A company automates processes using AI (dataset given).

Dataset

Process	Time Saved (%)	Error Reduction (%)	Cost Reduction (%)
Invoice Processing	60	70	50
Customer Support	50	40	30

Issues

Issue	Frequency
System Failure	Medium
Over-reliance	High

Recommend a reliable automation strategy.

40 A company uses AI across operations (dataset given).

Pipeline Data

Stage	Accuracy (%)	Human Involvement (%)
Forecasting	80	30
Decision Making	75	50
Execution	85	20

Issues

Issue	Impact
Bias in Predictions	High
Lack of Transparency	High

Evaluate performance and discuss a trustworthy framework.

UNIT 3 : Generative AI in Science and Research

Sr. No.	Question
1	A pharmaceutical startup has a dataset of known drug compounds. Apply a generative AI technique to generate new candidate molecules for further testing.
2	A climate research intern is given limited temperature data for a region. Demonstrate the use of generative AI to create additional synthetic climate records.
3	An ecology student needs to study population growth patterns but lacks sufficient data. Use a generative model to simulate basic population scenarios.
4	A pharmaceutical company uses generative AI to analyze and simulate molecular interactions for drug discovery. (a) Demonstrate the use of generative AI in modeling molecular interactions for drug discovery. (b) Examine how generative AI improves the efficiency and outcomes of the drug discovery process.
5	A biomedical lab uses GANs to generate synthetic MRI brain scan images for detecting tumors due to limited real patient data.Illustrate the role of GAN-generated MRI brain scans in improving the training dataset for tumor detection models.

6	A global research team works across different time zones and is willing to use digital platforms to collaborate on research projects and report writing. (a) Recommend suitable generative AI tool assist in creating and editing collaborative research reports. (b) Illustrate the importance of collaborative work in global research teams using generative AI tools.
7	A material science lab wants variations of a chemical compound. Apply generative AI to create multiple compound structures.
8	A research team needs to quickly share findings. Use generative AI to summarize and distribute knowledge among members.
9	A biotech company is working on vaccine development. Apply generative AI to generate new protein structures using existing datasets.
10	A biomedical lab uses GANs to generate synthetic MRI brain scan images for detecting tumors due to limited real patient data.Illustrate the role of GAN-generated MRI brain scans in improving the training dataset for tumor detection models.
11	A multidisciplinary research project requires collaboration among experts from different domains. Using Generative AI, suggest appropriate Generative AI tools that can facilitate interdisciplinary research activities such as knowledge integration, content creation, and collaborative communication. Explain your answer with suitable examples.
12	A healthcare organization wants to personalize drug development based on patient-specific genetic information. Using Generative AI, recommend how generative models can be used to design personalized drugs and simulate their outcomes before clinical trials. Justify your approach.
13	A financial analyst needs to evaluate risks under uncertain conditions. Apply generative AI to simulate financial market trends.
14	A research lab is exploring alternative experimental setups. Implement generative AI to generate multiple experiment configurations.
15	A global research team works across time zones. Recommend generative AI tools to create collaborative research reports.
16	An autonomous vehicle company needs diverse training data. Apply generative AI to generate realistic driving scenarios.
17	An engineering team is testing new materials. Demonstrate the use of generative AI to predict material properties.
18	A multilingual research group needs paper summaries. Implement generative AI to produce summaries in multiple languages.
19	Public health officials want to study disease spread. Apply generative AI to simulate infection scenarios under different conditions.

20	A dataset used in research has missing values. Use generative AI to generate and fill missing data points.
21	A physics researcher wants to test different experimental conditions. Demonstrate the use of generative AI to design experiment variations.
22	A microscopy lab has low-resolution images. Apply generative AI to enhance image quality for analysis.
23	An agriculture research center lacks sufficient crop data. Implement generative AI to create synthetic datasets for crop yield prediction.
24	A genomics researcher identifies unknown patterns. Use generative AI to generate multiple hypotheses for further study.
25	A climate scientist needs to study future weather conditions. Apply generative AI to simulate weather scenarios.
26	A research community platform is being developed. Demonstrate the use of generative AI for knowledge sharing among researchers.
27	An e-learning platform needs customized scientific content. Implement generative AI to generate personalized learning materials.
28	A pharmaceutical company is analyzing molecular interactions. Apply generative AI to model interactions in drug discovery.
29	A robotics lab needs diverse test environments. Use generative AI to generate realistic simulation scenarios.
30	A dataset shows unusual patterns. Demonstrate the use of generative AI to detect anomalies.
31	A rare disease study lacks sufficient patient data. Apply generative AI to generate synthetic datasets for research.
32	A multidisciplinary project requires collaboration.Recommend generative AI tools to support interdisciplinary research workflows.
33	A research organization aims to discover new drugs using genomic data. Apply generative AI to design a complete drug discovery pipeline integrating hypothesis generation and simulation.
34	Climate scientists need long-term predictions using diverse datasets. Implement a generative AI framework to simulate climate change impacts over decades.
35	A global research team works across different time zones and is willing to use digital platforms to collaborate on research projects and report writing. (a) Recommend suitable generative AI tool assist in creating and editing collaborative research reports. (b) Illustrate the importance of collaborative work in global research teams using generative AI tools..

UNIT : 4 Responsible AI

🟢 Easy Level (20% – 8 Questions)

Sr. No	Question
1	Identify key components of Responsible AI.
2	List ethical principles applicable to AI systems.
3	Recognize examples of algorithmic bias in AI applications.
4	A hiring AI system is trained using historical employee data where most of the selected candidates are male. The model starts favoring male candidates over equally qualified female candidates. → Identify and classify the type(s) of bias present in this dataset.
5	Illustrate transparency in AI with a simple example.
6	Summarize the concept of fairness in AI systems.
7	Identify responsibilities of AI developers.
8	Outline basic privacy concerns in AI applications.

Moderate Level (60% – 24 Questions)

Sr. No	Question
9	A hospital deploys an AI system to assist in diagnosing diseases. The system provides accurate results but does not explain how decisions are made, leading to distrust among doctors. → Analyze the scenario and explain the significance of Responsible AI in this context.
10	A loan approval AI system consistently rejects applications from certain neighborhoods due to historical financial data patterns. Analyze how such bias can affect economic equality and social mobility.
11	An AI-based grading system used in a school is observed to give higher scores to students from a particular background, raising concerns about fairness and bias. Analyze the situation and answer the following: 1. How can bias be identified by comparing average outcomes across different groups? 2. How does fairness evaluation contribute to improving the performance of AI systems? 3. How does removing bias enhance trust and reliability in AI-based grading systems?
12	A loan approval AI system consistently rejects applications from certain neighborhoods due to historical financial data patterns. Analyze how such bias can affect economic equality and social mobility.
13	Illustrate explainability in AI using practical scenarios.
14	A loan approval AI system consistently rejects applications from certain neighborhoods due to historical financial data patterns. → Analyze how such bias can affect economic equality and social mobility.
15	Apply techniques to reduce bias in datasets.
16	Demonstrate preprocessing methods for unbiased AI models.
17	Apply in-processing techniques to improve fairness.
18	Demonstrate post-processing methods for bias mitigation.
19	A social media company uses AI algorithms to recommend content. Users complain about misinformation, while the company focuses on engagement metrics. → Analyze the roles of platform owners, developers, users, and regulators in managing such issues.
20	Analyze responsibilities of developers in ethical AI practices.
21	Interpret regulatory roles in governing AI systems.
22	Analyze ethical challenges in Generative AI systems.

23	Examine sources through which bias enters AI models.
24	An AI-based grading system gives consistently higher scores to students from a particular background. → Apply fairness evaluation techniques to assess bias in grading outcomes.
25	A company develops an AI model for hiring using a dataset collected from past employee records. Over time, it is observed that the system favors certain groups while disadvantaging others. → Analyze how the characteristics of the dataset influence fairness in the AI system.
26	A healthcare organization deploys an AI system to assist doctors in diagnosing diseases, but final decisions are always reviewed by medical experts before implementation. → Illustrate how human-in-the-loop mechanisms function in this system and their importance.
27	A mobile application collects user data to provide personalized recommendations, but users are unaware of how their personal information is stored and used. A data breach later exposes sensitive information. → Examine the privacy risks associated with this AI system and their potential consequences.
28	A financial institution uses an AI-based system to automatically approve or reject loan applications without human intervention. Some applicants are denied without clear explanations. → Analyze the ethical concerns arising from automated decision-making in this scenario.
29	An AI model used for disease prediction is developed without proper clinical validation, leading to unreliable outputs. Propose strategies to ensure robustness, validation, and regulatory compliance.
30	An AI-based grading system used in a school is observed to give higher scores to students from a particular background, raising concerns about fairness and bias. Analyze the situation and answer the following: **1.** How can bias be identified by comparing average outcomes across different groups? **2.** How does fairness evaluation contribute to improving the performance of AI systems? **3.** How does removing bias enhance trust and reliability in AI-based grading systems?
31	Analyze risks associated with biased Generative AI outputs.
32	Differentiate fairness and equity in AI systems.

Hard Level (20% – 8 Questions)

Sr. No	Question
33	A multinational company develops an AI-based hiring system and adopts a single ethical guideline focused only on accuracy and efficiency. Over time, the system begins to show unfair hiring patterns across different demographic groups. → Evaluate the suitability of the ethical framework used and assess how a more comprehensive framework could improve outcomes.
34	A well-known tech company deploys an AI-powered facial recognition system that later shows high error rates for certain communities, leading to public criticism and regulatory action. → Analyze this real-world failure and highlight the role of bias in causing these issues along with its consequences.
35	A startup plans to build an AI system for loan approval that must ensure fairness, transparency, and accountability from the beginning. → Design a structured framework for developing an unbiased AI system, considering all stages from data collection to deployment.
36	An AI model used in healthcare shows biased predictions due to imbalanced training data. The development team considers multiple bias mitigation approaches at different stages of the pipeline. → Evaluate various bias mitigation techniques and justify the most effective approach for this scenario with suitable examples.
37	An AI model used for disease prediction is developed without proper clinical validation, leading to unreliable outputs. → Propose strategies to ensure robustness, validation, and regulatory compliance.
38	A hiring AI system is trained using historical employee data where most of the selected candidates are male. The model starts favoring male candidates over equally qualified female candidates. Identify and classify the type(s) of bias present in this dataset.
39	A hospital deploys an AI system to assist in diagnosing diseases. The system provides accurate results but does not explain how decisions are made, leading to distrust among doctors. Analyze the scenario and explain the significance of Responsible AI in this context.
40	Evaluate the working and applications of IBM AI Fairness 360 Toolkit.